

Leveraging Pre-Trained Models for Automated Meme Generation

Group 33:

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Abstract

This paper explores co-creative systems, emphasizing the design and implementation of interactive creative processes. We developed two distinct approaches for meme generation using pre-trained models. The first approach utilizes the GPT-2 model along with an input image to generate memes with captions based on a pre-defined prompt. The second approach adopts a supervised learning method, where a widely used pre-trained model from Hugging Face is fine-tuned on a new dataset consisting of images and corresponding labels. This process adapts the model to generate humorous captions for previously unseen meme templates. By comparing these methods, we evaluate their performance based on creativity, interactivity and humor quality. Our findings highlight the strengths and limitations of each approach, offering insights into the challenges of automating meme creation and the potential for integrating pre-trained models into creative AI systems.

Introduction

Throughout history, humans have developed countless ways to communicate, adapting methods to suit the context and culture of their times. In the last decade, memes have emerged as a powerful and creative medium of communication, enabling people to share humor and commentary on socio-cultural topics with friends and family. Memes, often combining visuals and text, have become a cornerstone of internet culture, known for their ability to compactly convey complex ideas with a humorous twist.

Simultaneously, the field of artificial intelligence (AI) has seen unprecedented growth, unlocking new possibilities for creative applications. This intersection of AI and creativity raises intriguing questions about how AI systems can be employed as co-creative partners in processes traditionally dominated by human ingenuity. Among these is the challenge of automating meme creation—a task that requires not only technical proficiency but also an understanding of humor and context.

To explore this, we developed two distinct approaches for meme generation using pre-trained AI models. The first approach leverages the GPT-2 model, coupled with an input

image, to generate captions based on predefined prompts. The second approach utilizes a supervised learning method, concentrating on fine-tuning a pre-trained BLIP model, a highly regarded framework available on Hugging Face, to generate humorous captions for novel and previously unseen meme templates. By investigating these methods, we aim to evaluate the potential of AI as a co-creative tool and examine its effectiveness in generating content that resonates with human audiences.

This research seeks to address the following key questions:

- How do different generative models, such as GPT-2 and fine-tuned BLIP, support co-creativity in the context of meme generation, and what trade-offs do they present between efficiency and user collaboration?
- In what ways can user interaction and feedback influence the co-creative potential of generative systems and how does this dynamic vary between highly automated models and those allowing greater user input?

Data

This section discusses the datasets utilized for the two distinct approaches implemented to generate creative and humorous memes.

In the first approach, the GPT-2 model is used to generate a caption based on the image description provided by the user via direct upload. Thus, in this case, the data are provided by the user himself, who is free to upload the desired images in the most common formats (.jpg, .jpeg, .png).

The second approach is based on fine-tuning the pre-trained BLIP model (Bootstrapped Language-Image Pre-training) and uses the dataset available in (Javaid 2025), which contains a large set of labeled meme images. It consists of 6992 images stored in compressed format, each labeled with specific information, such as category (satirical, humorous, political, etc.) or type of associated emotion (happiness, sadness, anger, etc.).

Methodology

As mentioned earlier we developed two approaches for meme generation with the integration of pre-trained mod-

els, focusing primarily on assisting in the caption generation process. The descriptions of these two approaches are as follows.

Caption created using GPT-2 model

We began by developing a user interface with *Streamlit*, providing an interactive platform where users can upload an input image. *Streamlit* was chosen to create an engaging experience, offering feedback to keep users informed and involved throughout the process.

Once the image is uploaded, it is processed using the BLIP (Bootstrapped Language-Image Pre-training) framework. Specifically, we utilized BLIP’s pre-trained processor and conditional generation model from the “*Salesforce/blip-image-captioning-base*” implementation to tokenize the image and generate a detailed text description of the uploaded input.

This generated description is then embedded into a custom prompt tailored to our requirements. For instance, in this case, the prompt instructs the model to “generate a short, funny, and sarcastic dark humor meme, keep it under 15 words, make it hilarious, and avoid links or irrelevant text.” The prompt can be adjusted as needed to align with various applications. Providing a more detailed prompt enhances the generation of accurate and task-specific captions.

To create the meme caption, we established a text-generation pipeline using the custom prompt as input along with some other additional parameters and specifying the use of the GPT-2 model. GPT-2 (Generative Pre-trained Transformer 2), developed by OpenAI, is a cutting-edge language model trained on a vast dataset of internet text. It excels in generating coherent and contextually relevant text by predicting the next word in a sentence, making it well-suited for crafting humorous and creative meme captions.

Finally, the generated caption is overlaid onto the input image with the assistance of the GPT-2 model. The final meme is then displayed to the user on the interactive interface created with *Streamlit*.

Caption created after training a pre-trained model on image dataset

In the second approach, we utilize a dataset comprising 6,992 images, which is pre-processed and subsequently divided into an 80% training set and a 20% validation set. The corresponding labels containing image captions are also loaded into dataframes for easy access and processing.

Similar to the first approach, this method involves converting raw image and text data into tokenized inputs compatible with the BLIP (Bootstrapped Language-Image Pre-training) framework.

However, in this case, the BLIP model is fine-tuned using the AdamW optimizer, which is commonly used for optimizing transformer-based models, and the model is trained on the training set.

The training process is conducted over a fixed number of epochs—five epochs in this case. After completing training and evaluating the model, an input image is processed by the fine-tuned model, which generates a caption adhering to the specified constraints.

Results

In this section we analyze the results obtained with both models, finally proposing a comparison between the two.

Starting from the first approach, we can say that the GPT-2 model demonstrated a good ability to generate meaningful, fluent and grammatically correct sentences. After providing as input the image of SpongeBob with his hands raised and a rainbow above him, as shown in Figure 1, the model generated a meme with a description that has an attempt at a sarcastic tone, which fits the tone of memes. In this case, the sentence reflects the sarcastic tone describing a social problem, but we can still say that the generated sentence represents a rather generic context, which does not fully respect the visual context of SpongeBob.



Figure 1: Meme generated with the GPT-2 model, in which the text “*Be prepared to be hit with the same backlash and criticism from everyone on social...*” reflects sarcastic and dark humor.

Figure 2 shows a second example of a meme generated with the GPT-2 model. The input consisted of an image depicting a man sitting on a bench observing the surrounding landscape. Again, we can say that the generated text has a sarcastic social vein, but it does not turn out to be totally related to the visual context of the image.

Therefore, this model has limitations in generating memes that present a description that has a specific connection to the visual details of the images. This is a plausible result because the GPT-2 model, not being trained on a set of images and captions, relies only on the described prompt. Consequently, as it struggles to establish a direct correlation between the visual content and the generated caption, the only solution is to be able to model the result obtained based on a more detailed and specific description in the prompt, by the user.

As for the second approach, we can say that the fine-tuning allowed the model to adapt to the specific domain of memes, succeeding not only in producing text that is fluid, sarcastic, and grammatically correct, but also

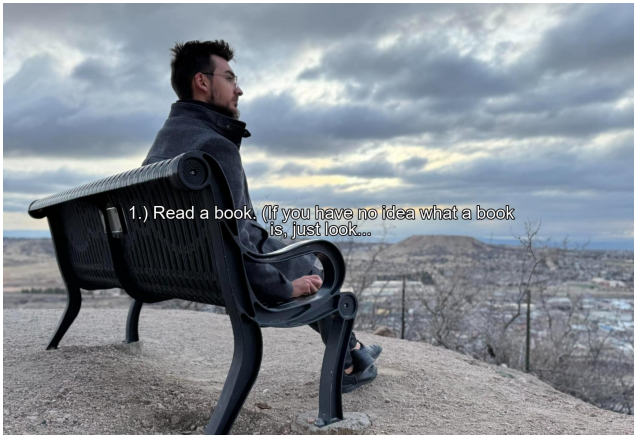


Figure 2: Memes generated using the GPT-2 model, in which the text “1.) Read a book. (If you have no idea what a book is, just look...” reflects an attempt at sarcastic humor.

visually relevant to the image. Figures 3 and 5 show the result obtained on two different examples of images before fine-tuning, while Figures 4 and 6 show, respectively, the progress achieved in meme generation on the same images after fine-tuning. From Figures 3 and 5, it can be said that, in the absence of fine-tuning, the model is unable to understand cultural references or exploit them creatively, and this actually leads to a result that is limited to the generation of text that is trivial, humor-poor or purely descriptive of the image itself. In fact, the text generated in Figure 3 “LEONARDO DIOGI IN THE GREAT GA GA GA GA...” merely creates text that attempts to be humorous, but is random and has no direct connection to the image. Similarly, the text generated in Figure 5 “a woman and a cat are eating food” turns out to be a literal visual description of the image, lacking humor or creativity. Fine tuning, on the other hand, allows for the generation of funny and engaging text that shows some relevance to the image. The text generated in Figure 4, “MR. BEAN AT WINE TASTING” exploits the ironic tone typical of memes, associating the image with Mr. Bean, known for his comical behavior, while maintaining the ‘wine’ factor present in the original image. Similarly, in Figure 6, the generated text “when you’re having a nice shift then someone else is furious” exploits the emotional dynamic of the scene contrasting the calm cat with the furious woman, adding a humorous tone.

The results show that even with some limitations on the part of both approaches, the implemented generative systems prove to be satisfactory, succeeding in their intent to create memes. To get a more complete view of the analysis of the results, we opted for a specific evaluation process, which gives us a better understanding of what both approaches actually achieved.

Evaluation

As a method of evaluating our generative systems, we decided to use a social media platform that would provide us

with a diverse group of users with different backgrounds, a wide age range, and interests in different fields. In addition, we wanted a tool that would allow us to collect feedback in a direct, authentic, and engaging way through interactions, comparisons, and comments. The platform that best met these needs turned out to be Instagram.

We therefore created a dedicated page (thedeepmemelab) to present the memes generated by our two systems (GPT-2 and BLIP fine-tuned), without revealing which model was responsible for each meme. After reaching a significant number of followers (around 30), we published the 10 most representative memes from our generative systems, 5 from each model. Instagram offers several evaluation tools, including like counts for each post, the number and type of comments, and the ability to create polls. The latter tool proved particularly crucial for our work, as it allowed a direct comparison between memes generated by the two systems. Users were asked to vote by expressing a preference, indicating which meme they found funniest. To make the process more engaging, we subsequently asked questions such as “What is your mood of the day?” or “Which one represents you the most?” The surveys provide results in the form of percentages indicating a preference for one of the two memes. This data, along with the number of “likes” each post received and the number and quality of comments (positive or negative), allowed us to conduct a comprehensive evaluation of our systems. The results obtained indicate which system was more engaging and creative in the eyes of the users. The results of this analysis are shown in Table 1.

From the Table 1, it can be stated that the fine-tuned BLIP appears to be the most effective model for meme generation, as it obtained more likes and more comments, especially positive comments, than the GPT-2 model. These results mirror those of the technical analysis reported above. These results mirror those of the technical analysis reported above. However, both generative systems achieved user engagement.

Discussion

The main goal of our work was to be able to implement a model that can collaborate with users in the creative process, combining human capabilities with those of the generative system and thus meeting the definition of co-creativity. Our evaluation and results highlight how both models, GPT-2 and BLIP with fine-tuning, contribute differently to this process, each offering unique strengths and limitations.

Although from the results obtained, the BLIP model with fine-tuning, being trained on a specific set of images and captions, appears to be better at understanding visual context and generating relevant text, at the same time it offers less influence on the user. In fact, this model appears to have a high degree of automation that limits the user’s ability to directly direct creative direction. Therefore, in this case, the user turns out to be more of an observer, who helps in the process of evaluating the meme by providing feedback that can highlight the progress of the generative system from the



Figure 3: Meme generated by the BLIP model **without** fine-tuning.



Figure 4: Meme generated by the BLIP model after fine-tuning.

a woman and a cat are eating food



Figure 5: Meme generated by the BLIP model **without** fine-tuning.

when you 're having a nice shift then someone else is furious



Figure 6: Meme generated by the BLIP model after fine-tuning.

Metric	Description	Result
Average number of likes (GPT-2 memes)	Average likes received for GPT-2 memes.	10
Average number of likes (BLIP fine-tuned memes)	Average likes received for BLIP fine-tuned memes.	13
Preferences in polls (GPT-2)	Average percentage of preferences in polls for GPT-2 memes.	35%
Preferences in polls (BLIP fine-tuned)	Average percentage of preferences in polls for BLIP fine-tuned memes.	65%
Total number of comments (GPT-2)	Total number of comments received on GPT-2 posts (5 memes).	2
Total number of comments (BLIP fine-tuned)	Total number of comments received on BLIP fine-tuned posts (5 memes).	6
Positive comments (GPT-2)	Total number of positive comments on GPT-2 memes.	1
Positive comments (BLIP fine-tuned)	Total number of positive comments on BLIP fine-tuned memes.	5
Negative comments (GPT-2)	Total number of negative comments on GPT-2 memes.	1
Negative comments (BLIP fine-tuned)	Total number of negative comments on BLIP fine-tuned memes.	1

Table 1: Results of the Instagram survey with around 30 users and 8 memes (4 per model).

perspective of creativity, degree of enjoyment, and correlation with the proposed image.

On the other hand, the GPT-2 model shows significant strengths from the perspective of user-system collaboration in the creative process. Indeed, this model allows the user to directly influence the prompt by giving specific directions to the system to generate the meme. This interaction allows users to incorporate creative intentions into the process,

guiding the tone, style and humor of the meme. This feature allows the system to increase the creativity of the memes generated by following precisely a particular guide chosen by the user, who can then range across topics.

Thus, we can say that the models represent distinct approaches to co-creativity, each with advantages and trade-offs: the fine-tuned BLIP model is more efficient in meme creation, but less collaborative in the creation process

itself. The GPT-2 model has more limitations in meme creation, but has a high level of user-system interaction.

Finally, we can say that the evaluation process we decided to undertake turns out to be highly embedded in the co-creative process. Nowadays there are numerous Instagram pages that publish funny memes, and some turn out to be very successful. The fact that our memes are directly evaluated by a diverse audience is itself part of the creative process. Instagram's digital communities are a hub of collective creativity, where content is constantly being created, shared and evaluated by a diverse audience. Expressing preferences, commenting on and sharing memes influences the evolution of the project and, quite spontaneously, has contributed to the evaluation of our generative systems, but it has also enriched our work with new and unexpected perspectives.

Future research

Based on our results and analysis, we can offer some insights to succeed in further improving the co-creativity of these types of generative systems, while maintaining a high level of quality of the generated memes. For example, some future project might succeed in the same intent through the use of hybrid systems, thus combining the flexibility of GPT-2 with BLIP's ability to understand visual context. This would provide a good balance from the standpoint of quality of results and high system-user interaction in the creation process itself.

Conclusion

In conclusion, our exploration of AI-driven meme generation using GPT-2 and fine-tuned BLIP model highlights the diverse roles generative systems can play in co-creative processes. While the BLIP model excels in efficiency and context-aware captioning, its high level of automation limits user collaboration. Conversely, the GPT-2 model fosters greater user-system interaction, enabling users to infuse their creative intentions into the process, though it may lack the same precision in meme creation. By integrating these approaches and leveraging feedback from digital communities, we have not only advanced our understanding of co-creativity but also demonstrated the potential of AI as a valuable partner in the creative process.

References

[Javaid 2025] Javaid, H. 2025. 6992 labeled meme images dataset. Available at <https://www.kaggle.com/datasets/hammadjavaid/6992-labeled-meme-images-dataset>.